

Juvenile myoclonic epilepsy refractory to treatment in a tertiary referral center

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ABSTRACT

Introduction: Juvenile myoclonic epilepsy (JME) is an epileptic syndrome often regarded as one in which seizures are relatively easy to control. Individuals with JME, however, often require lifelong therapy to remain seizure-free, and a few have refractory epilepsy. We ascertained a population with JME and characterized a subgroup with refractory epilepsy.

Material and methods: We audited and reviewed clinical records of individuals diagnosed with JME identified via a sample of 6600 individuals in a clinical database from a specialized epilepsy clinic at a tertiary referral center. **Results:** We identified 240 people with a diagnosis of JME (146 females), with a mean age at seizure onset of 14.2 years (SD: 4.5), and a mean age at diagnosis of 15.6 years (SD: 4.9). Clinical phenotypes seen were classic JME phenotype (88%), childhood absence epilepsy evolving into JME (6%), JME with adolescent absences (4%), and JME with astatic seizures (2%). More than a quarter (28%) had a family history of epilepsy. The most commonly used antiepileptic drug (AED) was sodium valproate in 78% of individuals, followed by levetiracetam (64%) and lamotrigine (55%). In the previous year, 47.5% were seizure-free. Using the International League against Epilepsy (ILAE) definitions and considering National Institute for Health and Care Excellence (NICE)-recommended AEDs for this syndrome, 121 individuals (50.4%) were identified as having refractory epilepsy.

Discussion: Juvenile myoclonic epilepsy is often regarded as a benign epileptic syndrome, but in this setting, half of the individuals with JME have refractory epilepsy with only about a quarter of those seizure-free in the previous year. Despite some advances in the understanding of this syndrome, there is still much to do before we can offer all the best outcomes.

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1. Introduction

The International League against Epilepsy (ILAE) [1] fully recognized juvenile myoclonic epilepsy (JME) in 1985, almost 120 years after the first clinical report [2] and 30 years after it was described [3]. Juvenile myoclonic epilepsy is defined by the ILAE [4] as an epilepsy syndrome, a subgroup of genetic generalized epilepsies (GGE), characterized by myoclonic jerks (MJ), which occur in a fully conscious state and predominate in the upper limbs, usually on awakening. Absence and generalized tonic-clonic seizures (GTCS) may occur in 50–80% of individuals. In 2013, an international experts panel proposed two sets of criteria for JME diagnosis [5], both relying on a clear history of myoclonic jerks predominantly after awakening and an electroencephalograph (EEG) with

generalized epileptiform discharges supporting a diagnosis of idiopathic generalized epilepsy (Box 1). Clinical phenotypes in JME were reported in the first descriptions [6–8], but only later [9] was a classification proposed with four subgroups: classic JME, childhood absence epilepsy (CAE) evolving to JME, JME with adolescent absence, and JME with astatic seizures (see Box 1).

Seizures in JME are often regarded as relatively easy to control. However, individuals frequently require lifelong therapy to remain seizure-free. A few present with refractory epilepsy, defined by the ILAE as failure of adequate trials of two tolerated and appropriately chosen and used antiepileptic drug (AED) schedules (whether as monotherapies or in combination) to attain sustained seizure freedom [10]. The UK National Institute for Health and Care Excellence (NICE) guidelines on the diagnosis and management of epilepsies [11] included recommendations for the pharmacological treatment of JME. Sodium valproate was recommended as first-line treatment, and lamotrigine, levetiracetam, and topiramate were considered first-line if sodium valproate was unsuitable or not tolerable. If first-line treatments were

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Box 1

Juvenile myoclonic epilepsy (JME) diagnostic criteria and clinical phenotypes.

Diagnostic criteria for JME (adapted from Kasteleijn-Nolst Trenité DG et al, <i>Epilepsy Behav</i> 2013 ^[5])	
Class I	1. Myoclonic jerks without loss of consciousness repeatedly occurring on awakening, i.e., within 2 h after awakening 2. EEG (routine, sleep, or sleep deprivation) that shows normal background and ictal generalized high amplitude polyspikes (and waves) with concomitant myoclonic jerks 3. Normal intelligence 4. Age at onset of between 10 and 25 years
Class II	1. Myoclonic jerks predominantly occurring on awakening 2. Myoclonic jerks facilitated by sleep deprivation and stress and provoked by visual stimuli and praxis or GTCs preceded by myoclonic jerks 3. EEG shows a normal background and at least once interictal generalized spike or poly-spike and waves with some asymmetry allowed with or without myoclonic jerks 4. No mental retardation or deterioration 5. Age at onset of between 6 and 25 years
JME clinical phenotypes (adapted from Martínez-Juárez IE et al, <i>Brain</i> 2005 ^[9])	
Classic JME	Adolescence onset of myoclonic, tonic-clonic and clonic-tonic-clonic seizures with or without rare-to-infrequent absences and an EEG with 4–6 Hz polyspike-wave complexes
CAE evolving to JME	Onset with absences with 3–4 Hz spike and wave complexes before aged 12 and then developed JME
JME with adolescent absence	Onset with absences with 3–5 Hz spike and polyspike and wave complexes aged 12 or older mixed with JME
JME with astatic seizures	Astatic seizures mixed with JME

ineffective or not tolerated, lamotrigine, levetiracetam, sodium valproate, or topiramate should be offered as adjunctive treatment. If adjunctive treatment is ineffective or not tolerated, clobazam, clonazepam, or zonisamide should be considered. Lastly, the guidelines emphasize that the use of carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, tiagabine, and vigabatrin should be avoided as they may aggravate seizures.

We audited a large subsample of people with epilepsy at a tertiary epilepsy center to identify those with JME with a special focus on the treatment used and to assess adherence to the NICE guidelines.

2. Material and methods

We performed a registered clinical audit to identify individuals with JME and to assess adherence to NICE guidelines regarding their treatment. We identified 1078 people with GGE from a subsample of 6600 individuals with epilepsy using an outpatient registry from the epilepsy clinic at the National Hospital for Neurology and Neurosurgery in London and the Chalfont Center in Chalfont St. Peter. We then reviewed individual records to identify those with JME according to the ILAE definition [4]. Those who had failed two appropriate AEDs on the grounds of efficacy were considered refractory. Clinical and demographic data collected included sex, age at seizure onset, age at diagnosis, coexisting seizure types, seizure facilitators, clinical phenotypes (according to the 2006 [9] classification), past medical history, and family history of epilepsy. Where available, EEG findings, brain imaging (magnetic resonance imaging [MRI]), and neuropsychological and psychiatric evaluations were reviewed. Only one EEG was considered per individual; whenever more than one was present, we considered the one most supportive of the diagnosis. Intermittent photostimulation was performed in all, but only few had sleep-deprived recordings. Current and past treatments were reviewed along with seizure control over the previous years, allowing for identification of people with a refractory seizures.

Data processing and statistical analysis were performed with Microsoft Excel 2013® and IBM SPSS (Statistical Package for the Social Sciences) V22.0®. Demographic and clinical characteristics were compared between the group with refractory seizures and the group with nonrefractory seizures using the 2-tailed, Wilcoxon rank-sum test for continuous variables with a non-normal distribution and the Fisher exact test for categorical variables. A *p* value of <0.05 was considered as statistically significant. No correction for multiple comparisons was performed as these were not considered in view of the small number of multiple comparisons.

3. Results

We identified 1078 people diagnosed with GGE, 240 of whom met the criteria for JME and were included. Demographic and clinical characteristics are summarized in Table 1. Using the ILAE definition and NICE-recommended AEDs for this syndrome, 121 individuals (50.4%) were identified as having refractory epilepsy. Table 2 characterizes and compares the group with refractory seizures and the group with nonrefractory seizures. The difference between the mean age at seizure onset and at diagnosis suggests a similar diagnostic delay of 1.4 years in both groups. Seizure types coexisting with myoclonus were convulsive seizures in 237 people (99%), absences in 98 (41%), and astatic seizures in 5 (2%).

Clinical phenotypes are described in Tables 1 and 2, with the majority (88%) having classic JME, especially in the group with nonrefractory seizures (96%). Other phenotypes were more prevalent in the group with refractory seizures. Sleep deprivation was described as a seizure precipitant by 102 people (43%), alcohol consumption by 44 (18%), photosensitivity by 33 (14%), and stress by 41 (17%). A family history of epilepsy was present in 67 people (28%), including in 38 first-degree relatives, 3 of which had JME.

Typical EEG findings were found in 192 people (80%), with photosensitivity present in 39 (20%). Normal EEG results were recorded in

Table 1
Clinical and demographic characteristics.

Number of patients	240
M:F ratio	94:146
Current age (y)	38 (SD 11.7)
Age at sz onset (y)	14.2 (SD 4.5)
Age at diagnosis (y)	15.6 (SD 4.9)
Clinical phenotype	
Classic	212 (88%)
CAE evolving into JME	14 (6%)
JME with adolescent Abs	9 (4%)
JME with astatic sz	5 (2%)
Past medical history	
Febrile convulsions	10 (4%)
Delayed language development	5 (2%)
Prematurity	5 (2%)
Asperger's syndrome	3 (1%)
Type 1 DM	4 (2%)
Nonrelevant	175 (73%)
Family history of epilepsy	
JME in first degree	3 (1%)
Non-JME in first degree	35 (15%)
JME in other members	3 (1%)
Non-JME in other members	26 (11%)

M — male; F — female; y — years; sz — seizures; CAE — childhood absence epilepsy; JME — juvenile myoclonic epilepsy; Abs — absences; DM — diabetes mellitus.

39 (16%) individuals, all of them on AED. Electroencephalograph was not available in nine (done elsewhere). Magnetic resonance imaging brain scans were performed in 216 people (90%) and were normal in 179 (85%). For the remaining, 15 showed nonspecific white matter changes, and in the others, incidental findings, including the following: diffuse brain atrophy in 3, cerebellar atrophy in 3, arachnoid cyst in 2, developmental venous anomaly in 2, and pineal cyst in 1.

Neuropsychological evaluation was available in 75 individuals (31%), showing mild widespread dysfunction in 15 (20%), executive dysfunction in 12 (16%), verbal learning dysfunction in 11 (14.7%), verbal and nonverbal memory weakness in seven (9.3%), visual memory dysfunction in three (4%), and normal results in 27 (36%). We did not correlate cognitive performance and AEDs in use. Thirty people (12.5%) had psychiatric evaluation, which suggested anxiety in 14 (47%), depression in 12 (40%), and normal results in four (13%).

The most commonly used AED over the recorded course of the condition was sodium valproate (188 people; 78%), followed by levetiracetam (154; 64.2%), lamotrigine (133; 55%), clobazam (63; 26%), carbamazepine (62; 26%), zonisamide (59; 25%), topiramate (54; 23%), phenytoin (38; 16%), clonazepam (36; 15%), phenobarbital (21; 9%), and ethosuximide (15; 6%). Other AEDs used by less than 15 individuals in total included lacosamide, perampanel, gabapentin, primidone, acetazolamide, vigabatrin, pregabalin, and sulthiame. Fig. 1 shows the total number of each AEDs ever prescribed, while Fig. 2 only presents the currently used AED. Two individuals had never taken AEDs, and six have withdrawn them because of poor seizure control. Seizure freedom with the first AED was seen in 41 (17%) individuals and in another 43 (18%) with the second AED. Usually as adjunctive drugs, NICE nonrecommended AEDs were used by 43 (18%) individuals. Forty-four of the 146 females were currently using sodium valproate (25 of childbearing age), and 60 had used this in the past. Amongst those on valproate, it was used as monotherapy in four and as part of a two-AED regimen in eleven, and in the remaining, it was used as part of a polytherapy regimen mainly due to refractoriness.

In the previous year, 114 individuals (48%) were seizure-free, 30 (13%) had only MJ or absences, and 84 (35%) also had convulsions. One died and 11 were lost to follow-up. The average follow-up was 21.8 years (range 1–60 years). On last follow-up, the majority (63%) of the individuals with nonrefractory seizures were taking one AED, while 86% of the individuals with refractory seizures were taking two or more AEDs.

Table 2
Differences of the group with refractory seizures and group with nonrefractory seizures.

	Group with refractory seizures	Group with nonrefractory seizures	p-Value
Number of patients	121	119	
M:F ratio	41:80	53:66	0.11
Current age (y)	38 (SD 10.6)	39 (SD 12.6)	0.75
Age at sz onset (y)	14 (SD 4.7)	15 (SD 4.1)	<0.05
Age at diagnosis (y)	15 (SD 5.1)	16 (SD 4.7)	<0.05
Clinical phenotype			<0.05
Classic	98 (81%)	114 (96%)	
CAE evolving into JME	12 (10%)	2 (2%)	
JME with adolescent Abs	6 (5%)	3 (2%)	
JME with astatic sz	5 (4%)	0 (0%)	
Past medical history			0.07
Febrile convulsions	8 (7%)	2 (2%)	
Delayed language development	3 (2%)	2 (2%)	
Prematurity	2 (2%)	3 (2%)	
Asperger's syndrome	3 (2%)	0 (0%)	
Type 1 DM	0 (0%)	4 (3%)	
Nonrelevant	105 (87%)	108 (91%)	
Family history of epilepsy			0.98
JME in first degree	1 (1%)	2 (2%)	
Non-JME in first degree	17 (14%)	18 (15%)	
JME in other members	2 (2%)	1 (1%)	
Non-JME in other members	13 (11%)	13 (11%)	
EEG			0.06
Typical JME	102 (88%)	90 (78%)	
[photoparoxysmal response]	15 (15%)	24 (27%)	
Normal	14 (12%)	25 (22%)	
Not available	5	4	
MRI			0.70
Normal	97 (86%)	82 (84%)	
Abnormal	16 (14%)	16 (16%)	
Not available	8	21	
Neuropsychology			0.59
Normal	18 (33%)	9 (43%)	
Abnormal	36 (67%)	12 (57%)	
Not available	67	98	
Neuropsychiatry			<0.05
Normal	0 (0%)	4 (31%)	
Abnormal	17 (100%)	9 (69%)	
Not available	104	106	
Current AED used			<0.05
0	0 (0%)	8 (7%)	
1	16 (14%)	72 (63%)	
2	52 (46%)	35 (30%)	
>2	45 (40%)	0 (0%)	
Total	113	115	
Follow-up			<0.05
Sz free	30 (25%)	84 (71%)	
Only MJ or Abs	14 (11%)	16 (13%)	
GTCS +/- MJ or Abs	69 (57%)	15 (13%)	
Lost to follow-up or death	8 (7%)	4 (3%)	

M — male; F — female; y — years; sz — seizures; CAE — childhood absence epilepsy; JME — juvenile myoclonic epilepsy; Abs — absences; DM — diabetes mellitus; AED — antiepileptic drugs; MJ — myoclonic jerks; GTCS — generalized tonic-clonic seizures.

4. Discussion

Juvenile myoclonic epilepsy is often regarded as a benign epileptic syndrome, with some reports even suggesting that some individuals are seizure-free without AEDs [12]. A recent study reported that over 90% of individuals with JME were in remission [13]. In our sample, however, about half of individuals have refractory epilepsy, suggesting a possible referral bias as the setting is a major tertiary referral center, which is more likely to attract referral of more refractory cases.

Females were more prevalent in both groups, as previously reported [13]. Age of onset was slightly higher in the group with easy-to-control seizures, but diagnostic delay was similar, suggesting that the diagnosis in people with refractory disease does not pose a more difficult diagnostic challenge. In our sample, we found that, although the classic clinical

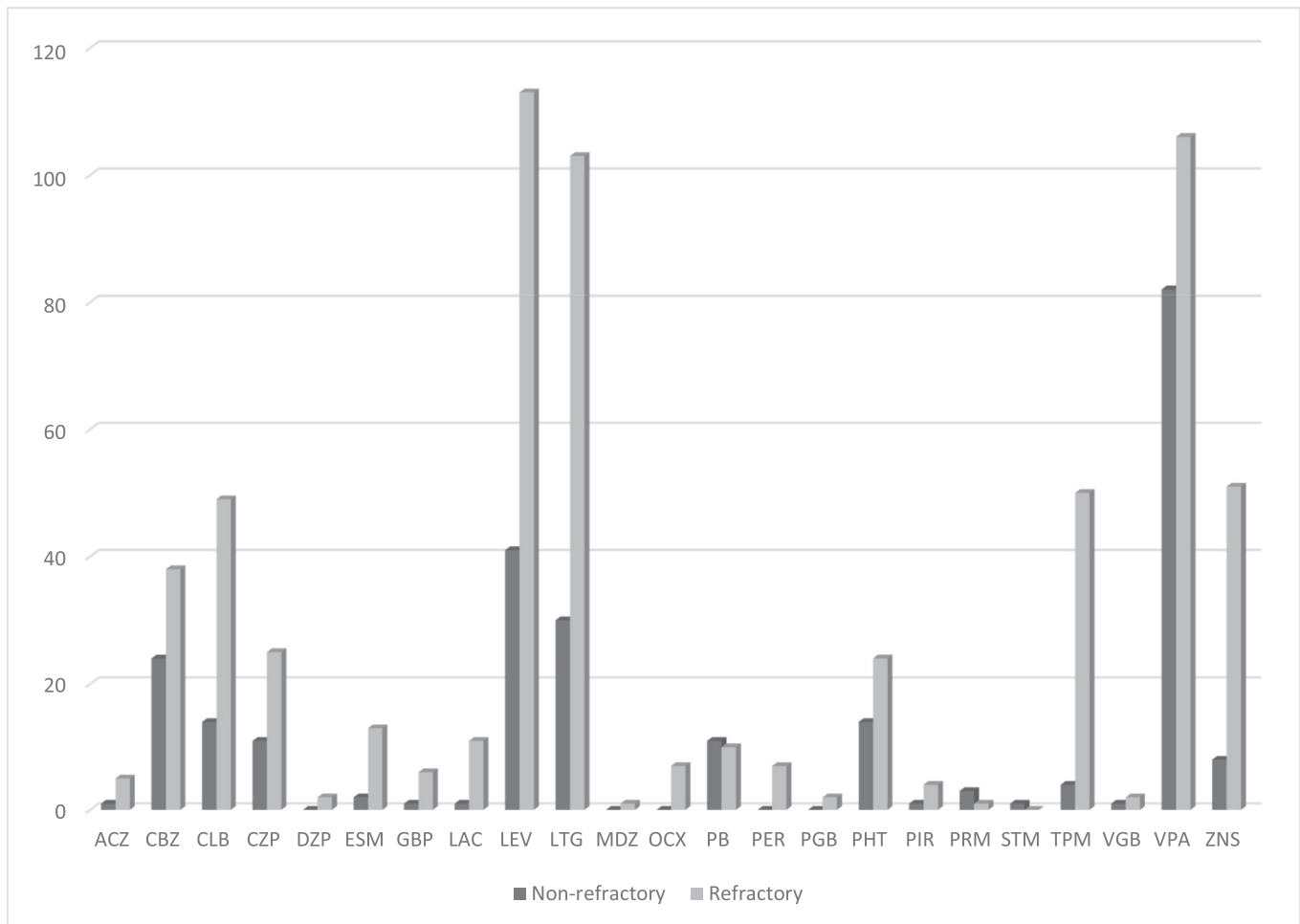


Fig. 1. Total AED used in the group with refractory seizures and the group with nonrefractory seizures. Legend: Total number of each AEDs ever prescribed in the group with refractory seizures and the group with nonrefractory seizures. ACZ – acetazolamide, CBZ – carbamazepine, CLB – clobazam, CZP – clonazepam, DZP – diazepam, ESM – ethosuximide, GBP – gabapentin, LAC – lacosamide, LEV – levetiracetam, LTG – lamotrigine, MDZ – midazolam, OXC – oxcarbazepine, PB – phenobarbital, PER – perampanel, PGB – pregabalin, PHT – phenytoin, PIR – piracetam, PRM – primidone, STM – sulthiame, TPM – topiramate, VGB – vigabatrin, VPA – valproic acid, ZNS – zonisamide.

phenotype was the most prevalent in both groups, other phenotype subgroups were more prevalent in the group with refractory seizures. The phenotype difference between groups was statistically significant. It was reported previously [9] that 72% of people presented with the classic subtype, with over half of these seizure-free, a fifth had CAE evolving to JME with 7% seizure-free, 7% had JME with adolescent absence with 56% seizure-free, and 3% had JME with astatic seizures with 62% seizure-free. Others also observed worse seizure control in those with previous CAE [14,15].

Only small nonsignificant differences were seen between groups regarding a family history of epilepsy. This was present in more than a quarter of individuals, but differences in the presence or absence of epilepsy in the family were not significant between the group with refractory seizures and the group with nonrefractory seizures. A similar percentage of relatives with epilepsy was formerly reported [13]. This supports previously reported studies suggesting an underlying genetic predisposition to JME [16–20].

Findings from EEGs were not different between groups. Abnormal MRI findings were interpreted as incidental, and group differences were not significant. Overall, results from imaging studies in JME are normal, but advanced neuroimaging studies suggest functional and structural abnormalities in the frontal cortex and thalamus [17,21]. This would be in accordance with some evidence of thalamofrontocortical network dysfunction [17,22–25] as a putative pathophysiological mechanism for JME. There seems to be a subnetwork of hyperconnectivity,

involving primary motor, supplementary motor, and the parietal and subcortical regions, which may contribute to myoclonic seizure generation [26–28]. The presence of altered motor system activation and functional connectivity in siblings unaffected by JME demonstrates that this is not medication- or seizure-related but may constitute a genetically determined JME endophenotype [17,19].

In our cohort, neuropsychological evaluation profiles often shown mild widespread malfunctioning and executive dysfunction, in concordance with previous reports that people with JME have executive and attention issues characterized by deficits in mental flexibility, planning, concept formation, inhibitory control, attentional processes, memory, visuospatial function, and verbal fluency [29–32]. In our sample, there were no group differences on neuropsychological evaluation. Neuropsychiatric evaluation, although infrequently performed, suggested that anxiety and depression are relatively common in these individuals, significantly more in the group with refractory seizures. Anxiety and depression were also the most common diagnosis in report in which about a fifth of people with JME had coexisting psychiatric comorbidities [13]. As not all individuals were evaluated and the impact of polymedication was not assessed, conclusions about neuropsychological and neuropsychiatric evaluations are challenging.

Significantly, more people in the group with refractory seizures are on polytherapy, and this should not come as a surprise. National Institute for Health and Care Excellence-recommended AEDs are amongst the most used in this cohort, as over three-quarters are using

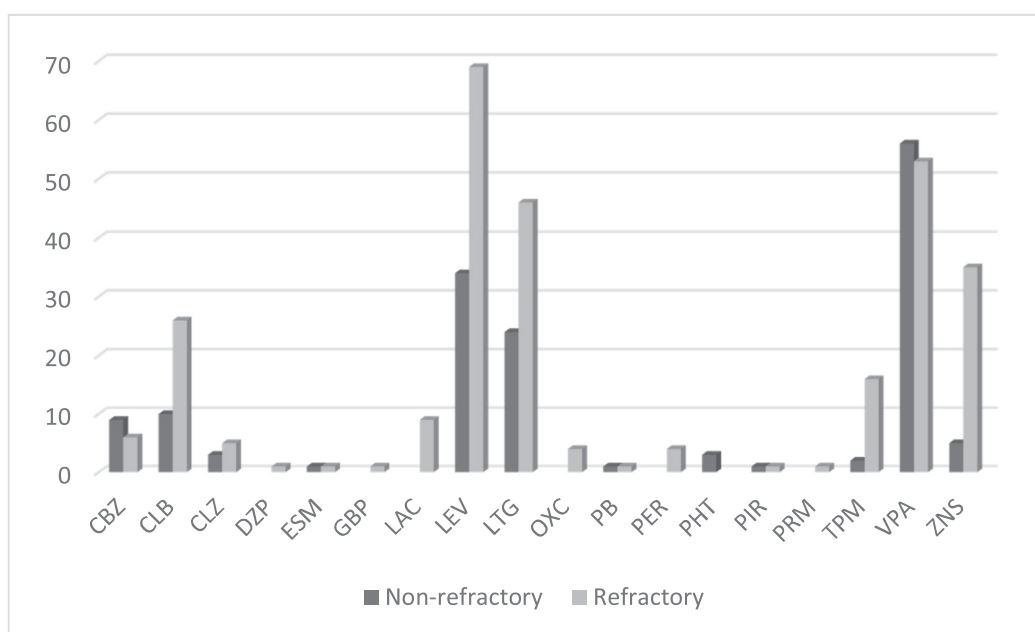


Fig. 2. Current AED used in the group with refractory seizures and the group with nonrefractory seizures. Legend: Total number of each AEDs currently prescribed in the group with refractory seizures and the group with nonrefractory seizures. ACZ – acetazolamide, CBZ – carbamazepine, CLB – clobazam, CLZ – clonazepam, DZP – diazepam, ESM – ethosuximide, GBP – gabapentin, LAC – lacosamide, LEV – levetiracetam, LTG – lamotrigine, MDZ – midazolam, OXC – oxcarbazepine, PB – phenobarbital, PER – perampanel, PGB – pregabalin, PHT – phenytoin, PIR – piracetam, PRM – primidone, STM – sulthiame, TPM – topiramate, VGB – vigabatrin, VPA – valproic acid, ZNS – zonisamide.

them. About a fifth were using AEDs outside the NICE recommendations. The SANAD study [33] compared valproate, lamotrigine, and topiramate in people with generalized seizures. The JME subgroup was not specifically addressed, but it was concluded that sodium valproate should remain as a first-line treatment for most people with idiopathic generalized epilepsy or seizures that are difficult to classify, whereas lamotrigine and topiramate should generally be avoided because of the inferior efficacy of lamotrigine and the inferior tolerability of topiramate [33]. Conversely, other studies support the use of topiramate [34] and lamotrigine [35] in people with JME. The most reasonable treatment path in these individuals is to follow the NICE guidelines. For women, in view of its potential teratogenicity, valproate should only be a last resort despite its efficacy in this syndrome [13].

Despite the type and number of AEDs used in different combinations, only a quarter of the group with refractory seizures were seizure-free in the previous year, compared with 71% of the group with nonrefractory seizures. Overall, 48% of individuals were seizure-free in the previous year, considerably lower than reported in other studies [12,36–39], which, as stated before, is probably due to the higher ratio of refractory epilepsy in our sample. Despite recent advances in the understanding of this syndrome, there is still much needed before we will be able to offer the best outcome to all as it has been shown that early seizure freedom improves social adjustment and occupational integration and is associated with a better quality of life [40].

A major limitation of this study is its retrospective nature and dependency on the accuracy and completeness of the existing medical records. Some people with JME amongst those diagnosed with GGE may have been missed. Seizure frequency data are often problematic in such retrospective studies, and this makes it a challenge to establish an exact correlation between AED changes and changes in seizure control, but we have attempted to deal with this in a pragmatic way. Adherence to AED was difficult to assess, and this could lead to some cases being “pseudorefractory”. The sample was also drawn from a population in a tertiary referral center, including incident and prevalent cases, which may have biased this study towards the more refractory end of the clinical spectrum of JME. The strength of this study is the size of the cohort with a thorough

comparison of the group with refractory conditions and the group with nonrefractory seizures, which adds information to the clinical ascertainment of this entity.

5. Conclusion

Juvenile myoclonic epilepsy is still a challenging condition, with variable prognosis and without clear markers to predict refractoriness. The challenge of seizure freedom is even greater in this syndrome than in other epilepsy syndromes because of the psychological profile of these individuals that may lead to difficulties in treatment adherence. Prospective multicenter studies of an incident population with refractory JME will probably bring more information on this common condition.

Disclosure

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References

- [1] Commission on Classification and Terminology of the International League against Epilepsy. Proposal for classification of epilepsies and epileptic syndromes. *Epilepsia* 1985;26:268–78.
- [2] Herpin T. Des acces incomplets d'épilepsie. Paris: Bailliere; 1867; 162–70.
- [3] Janz D, Christian W. Impulsiv petit mal. *Dtsch Z Nervenheilkd* 1957;176:346–86.
- [4] Engel Jr J, ILAE. A proposed diagnostic scheme for people with epileptic seizures and with epilepsy: report of the ILAE task force on classification and terminology. *Epilepsia* 2001;42:796–803.
- [5] Kasteleijn-Nolst Trenité DG, Schmitz B, Janz D, Delgado-Escueta AV, Thomas P, Hirsch E, et al. Consensus on diagnosis and management of JME: from founder's observations to current trends. *Epilepsy Behav* 2013;28(Suppl. 1):S87–90.
- [6] Janz D. Epilepsy with impulsive petit mal (juvenile myoclonic epilepsy). *Acta Neurol Scand* 1985;72:449–59.
- [7] Tsuboi T. Primary generalized epilepsy with sporadic myoclonias of myoclonic petit mal type. A clinical, electroencephalographic, statistical and genetic study of 399 probands. Stuttgart: Thieme; 1977.
- [8] Mai R, Canevini MP, Pontrelli V, Tassi L, Bertin C, Di Marco C, et al. L'epilessia mioclonica giovanile di Janz: analisi prospettica di un campione di 57 pazienti. *Boll Lega It Epil* 1990;70/71:307–9.
- [9] Martínez-Juárez IE, Alonso ME, Medina MT, Durón RM, Bailey JN, López-Ruiz M, et al. Juvenile myoclonic epilepsy subsyndromes: family studies and long-term follow-up. *Brain* 2006;129:1269–80.
- [10] Kwan P, Arzimanoglou A, Berg AT, Brodie MJ, Hauser WA, Mathern G, et al. Definition of drug resistant epilepsy. Consensus proposal by the ad hoc Task Force of the ILAE Commission on Therapeutic Strategies. *Epilepsia* 2010;51(6):1069–77.
- [11] National Institute for Health and Clinical Excellence. Epilepsies: diagnosis and management. NICE guidelines [CG137]. <https://www.nice.org.uk/guidance/cg137>; 2012. [accessed 12 October 2017].
- [12] Camfield CS, Camfield PR. Juvenile myoclonic epilepsy 25 years after seizure onset: a population-based study. *Neurology* 2009;73(13):1041–5.
- [13] Chowdhury A, Brodie MJ. Pharmacological outcomes in juvenile myoclonic epilepsy: support for sodium valproate. *Epilepsy Res* 2016;119:62–6.
- [14] Wirrell E, Camfield C, Camfield P, Dooley J. Prognostic significance of failure of the initial antiepileptic drug in children with absence epilepsy. *Epilepsia* 2001;42:760–3.
- [15] Guaranha MS, Filho GM, Lin K, Guilhoto LM, Caboclo LO, Yacubian EM. Prognosis of juvenile myoclonic epilepsy is related to endophenotypes. *Seizure* 2011;20:42–8.
- [16] Schmitz B, Yacubian EM, Feucht M, Hermann B, Trimble M. Neuropsychology and behavior in juvenile myoclonic epilepsy. *Epilepsy Behav* 2013;28(Suppl. 1):S72–3.
- [17] Wandschneider B, Thompson PJ, Vollmar C, Koepp MJ. Frontal lobe function and structure in juvenile myoclonic epilepsy: a comprehensive review of neuropsychological and imaging data. *Epilepsia* 2012;53:2091–8.
- [18] Iqbal N, Caswell H, Muir R, Cadden A, Ferguson S, Mackenzie H, et al. Neuropsychological profiles of patients with juvenile myoclonic epilepsy and their siblings: an extended study. *Epilepsia* 2015;56:1301–8.
- [19] Wandschneider B, Centeno M, Vollmar C, Symms M, Thompson PJ, Duncan JS, et al. Motor co-activation in siblings of patients with juvenile myoclonic epilepsy: an imaging endophenotype? *Brain* 2014;137:2469–79.
- [20] Cossette P. Channelopathies and juvenile myoclonic epilepsy. *Epilepsia* 2010;51(Suppl. 1):30–2.
- [21] Anderson J, Hamandi K. Understanding juvenile myoclonic epilepsy: contributions from neuroimaging. *Epilepsy Res* 2011;94:127–37.
- [22] Kim SH, Lim SC, Kim W, Kwon OH, Jeon S, Lee JM, et al. Extrafrontal structural changes in juvenile myoclonic epilepsy: a topographic analysis of combined structural and microstructural brain imaging. *Seizure* 2015;30:124–31.
- [23] Alhusaini S, Ronan L, Scanlon C, Whelan CD, Doherty CP, Delanty N, et al. Regional increase of cerebral cortex thickness in juvenile myoclonic epilepsy. *Epilepsia* 2013;54:e138–1.
- [24] Kim JB, Suh SI, Seo WK, Oh K, Koh SB, Kim JH. Altered thalamocortical functional connectivity in idiopathic generalized epilepsy. *Epilepsia* 2014;55:592–600.
- [25] Kim JH, Suh SI, Park SY, Seo WK, Koh I, Koh SB, et al. Microstructural white matter abnormality and frontal cognitive dysfunctions in juvenile myoclonic epilepsy. *Epilepsia* 2012;53:1371–8.
- [26] O'Muirheartaigh J, Vollmar C, Barker GJ, Kumari V, Symms MR, Thompson P, et al. Abnormal thalamocortical structural and functional connectivity in juvenile myoclonic epilepsy. *Brain* 2012;135:3635–44.
- [27] Focke NK, Diederich C, Helms G, Nitsche MA, Lerche H, Paulus W. Idiopathic-generalized epilepsy shows profound white matter diffusion-tensor imaging alterations. *Hum Brain Mapp* 2014;35:3332–42.
- [28] Caeyenberghs K, Powell HW, Thomas RH, Brindley L, Church C, Evans J, et al. Hyperconnectivity in juvenile myoclonic epilepsy: a network analysis. *Neuroimage Clin* 2014;7:98–104.
- [29] Seneviratne U, Cook M, D'Souza W. Focal abnormalities in idiopathic generalized epilepsy: a critical review of the literature. *Epilepsia* 2014;55:1157–69.
- [30] Loughman A, Bowden SC, D'Souza W. Cognitive functioning in idiopathic generalised epilepsies: a systematic review and meta-analysis. *Neurosci Biobehav Rev* 2014;43:20–34.
- [31] Valente KD, Rzezak P, Moschetta SP, de Vincentiis S, Coan AC, Guerreiro CA. Delineating behavioral and cognitive phenotypes in juvenile myoclonic epilepsy: are we missing the forest for the trees? *Epilepsy Behav* 2016;54:95–9.
- [32] Cevik N, Koksai A, Dogan VB, Dirican AC, Bayramoglu S, Ozturk M, et al. Evaluation of cognitive functions of juvenile myoclonic epileptic patients by magnetic resonance spectroscopy and neuropsychiatric cognitive tests concurrently. *Neurol Sci* 2016;37(4):623–7.
- [33] Marson AG, Al-Kharusi AM, Alwaidh M, Appleton R, Baker GA, Chadwick DW, et al. The SANAD study of effectiveness of valproate, lamotrigine, or topiramate for generalised and unclassifiable epilepsy: an unblinded randomised controlled trial. *Lancet* 2007;369:1016–26 [SANAD Study group].
- [34] Park KM, Kim SH, Nho SK, Shin KJ, Park J, Ha SY, et al. A randomized open-label observational study to compare the efficacy and tolerability between topiramate and valproate in juvenile myoclonic epilepsy. *J Clin Neurosci* 2013;20:1079–82.
- [35] Machado RA, García VF, Astencio AG, Cuartas VB. Efficacy and tolerability of lamotrigine in juvenile myoclonic epilepsy in adults: a prospective, unblinded randomized controlled trial. *Seizure* 2013;22:846–55.
- [36] Höfler J, Unterberger I, Dobesberger J, Kuchukhidze G, Walser G, Trinka E. Seizure outcome in 175 patients with juvenile myoclonic epilepsy—a long-term observational study. *Epilepsy Res* 2014;108:1817–24.
- [37] Baykan B, Altındag EA, Bebek N, Ozturk AY, Aslantas B, Gurses C, et al. Myoclonic seizures subside in the fourth decade in juvenile myoclonic epilepsy. *Neurology* 2008;70:2123–9.
- [38] Geithner J, Schneider F, Wang Z, Berneiser J, Herzer R, Kessler C, et al. Predictors for long-term seizure outcome in juvenile myoclonic epilepsy: 25–63 years of follow-up. *Epilepsia* 2012;53:1379–86.
- [39] Senf P, Schmitz B, Holtkamp M, Janz D. Prognosis of juvenile myoclonic epilepsy 45 years after onset: seizure outcome and predictors. *Neurology* 2013;81:2128–33.
- [40] Schneider-von Podewils F, Gasse C, Geithner J, Wang ZI, Bombach P, Berneiser J, et al. Clinical predictors of the long-term social outcome and quality of life in juvenile myoclonic epilepsy: 20–65 years of follow-up. *Epilepsia* 2014;55:322–30.